CNS Infections: Parasites & Fungi

Tony A. Slieman, M.Sc., Ph.D.
Director of Academic Affairs and Associate Professor
Founding Director of Master's in Biomedical Sciences
Department of Biomedical and Anatomical Sciences
tslieman@nyit.edu

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

1. Describe the different types of fungal and parasitic CNS infections and their etiology
2. Compare and contrast parasitic and fungal meningitis
3. Compare the properties of the parasitic and fungal etiological agents of meningitis, encephalitis, myelitis and brain abscesses
4. Identify the importance of diagnostic CSF analysis and antimicrobial therapy
5. Illustrate epidemiological factors in the determination of etiology

Source: Course Syllabus

Terms

- Meningitis (Acute, Aseptic & **Chronic**)
- **Encephalitis/Myelitis**
- **Brain abscesses**

CNS Infection (Chronic Meningitis)

- Chronic Meningitis
  - Meningeal inflammation with CSF pleocytosis > 4 weeks
  - Symptoms evolve over days/weeks
    a) Headache, fever, etc.
    b) Nuchal rigidity subtle/absent
    c) Progresses → seizures, mental status changes, confusion or hallucinations
    d) Focal neurologic deficits develop
    e) Hydrocephalus & increased intracranial pressure

Chronic meningitis etiological agents

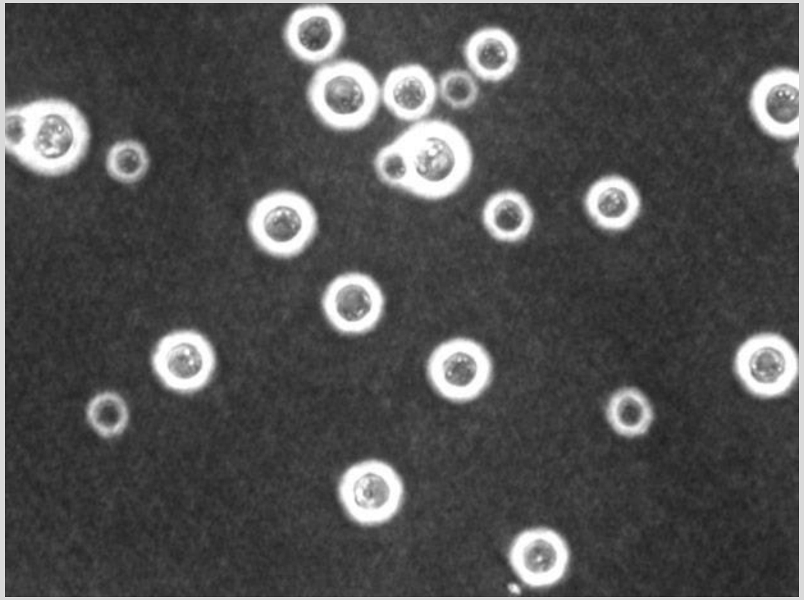
- Fungal:
  - *Aspergillus* spp.
  - *Candida* spp.
  - *Coccidioides* spp.
  - *Cryptococcus* spp.
  - *Histoplasma* spp.
  - *Pseudallescheria* spp.
  - *Sporothrix* spp.
- Parasites:
  - *Taenia solium*
  - *Angiostrongylus* spp.

Fungal Chronic Meningitis

- General themes:
  - Opportunistic
    a) Immune suppression
    b) Immune defects (CMI)
  - Moderate lymphocytic pleocytosis

Fungal Chronic Meningitis

- *Cryptococcus neoformans*
  - Presentation: subacute meningoencephalitis to fever of unknown origin, hydrocephalus
  - Laboratory:
    a) Cryptococcal antigen latex agglutination
    b) Microscopy of India ink preparations less sensitive (capsules)
  - Treatment: amphotericin B + flucytosine



http://sequence-www.stanford.edu/group/C.neoformans/images/cneoindiaink.jpg

Fungal Chronic Meningitis

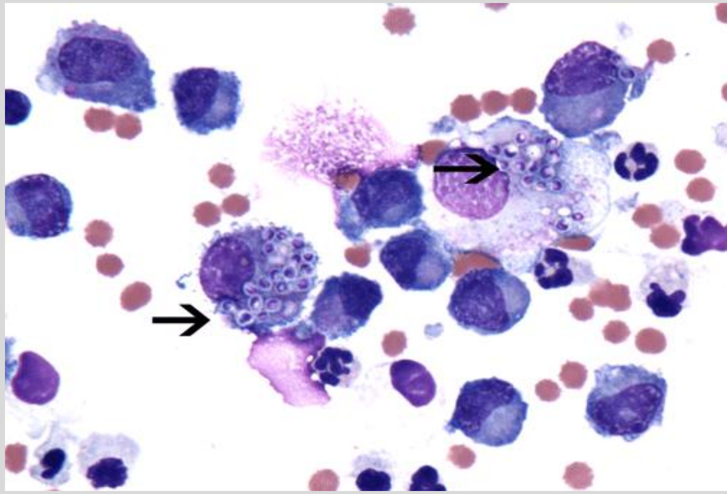
- *Coccidioides posadasii & C. immitis*
  - Presentation: erythema nodosum, hydrocephalus
  - Laboratory:
    - a) CSF eosinophilia (endemic)
    - b) Complement-fixing antibodies (75-95% of cases)
    - c) Culture (50% of cases; hazardous)
  - Treatment: amphotericin B + azole

http://www.pcds.org.uk/ee/images/made/ee/images/uploads/clinical/EN_13_640_491_70_http:www.pcds.org.ukeeassetsimgwatermark.gif_0_0_80_r_b_-5_-5_.jpg

Fungal Chronic Meningitis

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- *Histoplasma capsulatum*
  - Rare
  - Endemic regions – the Ohio River Valley
  - Presentation: fever, oral lesions, hepatosplenomegaly, hydrocephalus
  - Laboratory:
    a) Culture (27-65% of cases)
    b) *Histoplasma* polysaccharide antigen in the urine/blood/CSF (61% of patients)
    c) Visualization of yeast cells inside macrophages
  - Treatment: amphotericin B



Fungal Chronic Meningitis

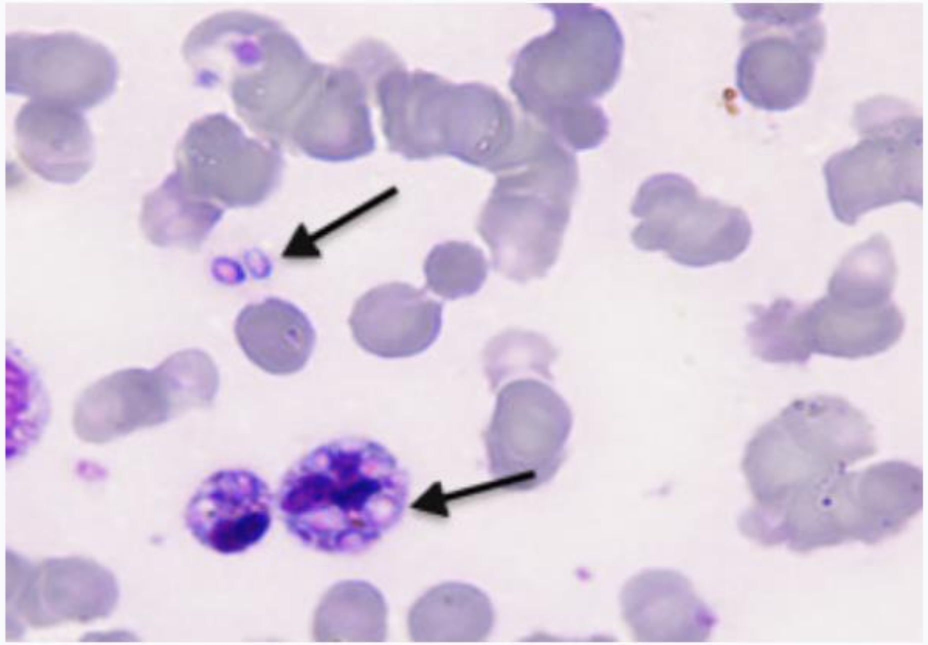
- *Pseudallescheria* spp. (*P. boydii)*
- Most common fungal etiologic agent of mycetoma
- Meningitis, meningoencephalitis & brain abscesses
- Most in immunocompromised (but seen in immunocompetent)
- Polluted water, sewage, swamps
- Near drowning victims
- Resistant to Amphotericin B (miconazole shows promise)

A 45-year-old male presents to his PCP with a fever, confusion and a severe headache. Microscopic analysis identified a fungal agent with a prominent capsule. Which of the following is the etiological agent of this disease?

- Coccidioides posadasii
- Histoplasma capsulatum
- Pseudallescheria boydii
- Cryptococcus neoformans

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

A 45 year old male presents to his PCP with a fever, confusion and a severe headache. The patient has recently went on a hiking trip exploring caverns. Microscopic analysis identified a fungal agent shown in the attached image. Which of the following is the etiological agent of this disease?



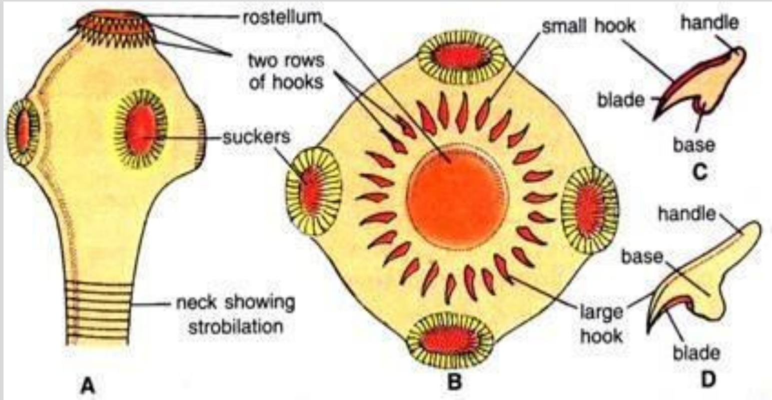
- Coccidioides posadasii
- Histoplasma capsulatum
- Pseudallescheria boydii
- Cryptococcus neoformans

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Parasitic Chronic Meningitis

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- *Taenia solium*
  - Pork tapeworm
  - Neurocysticercosis
  - Endemic: Mexico, South America & Asia
  - Presentation:
    - a) Seizures (most common)
    - b) Obstructive cysts (hydrocephalus)
    - c) Space-occupying lesions
  - Laboratory:
    - a) CT of head shows calcified legions
    - b) Lymphocytic pleocytosis with eosinophils
  - Treatment:
    - Albendazole, steroids, analgesics, antiepileptic drugs
    - Surgical resection of lesions & placement of ventricular shunts



http://cdn.biologydiscussion.com/wp-content/uploads/2016/04/clip_image007-14.jpg

https://grouphaty.wikispaces.com/file/view/Taenia-solium3.jpg/310782944/Taenia-solium3.jpg

https://web.stanford.edu/group/parasites/ParaSites2005/cysticercosis/parasite/taenia_egg_3.gif

Parasitic Chronic Meningitis

- *Angiostrongylus cantonensis*
  - rat lung worm (round worm)
  - Endemic: Asia and the Pacific Islands
  - Ingestion of raw/inadequately cooked shellfish or snails
  - Incubation period: can be up to 1 year
  - Presentation:
    a) Typical (nausea, vomiting, neck stiffness, & headaches that are global & severe)
    b) Rash with pruritus is also common
  - Diagnosis:
    a) History (area and food)
    b) CSF and peripheral eosinophilia
  - Treatment:
    a) None
    b) Resolve spontaneously over time

https://www.cdc.gov/dpdx/images/angiostrongyliasis/Angio2_dic_400x.jpg

Other rare causes of Chronic Meningitis

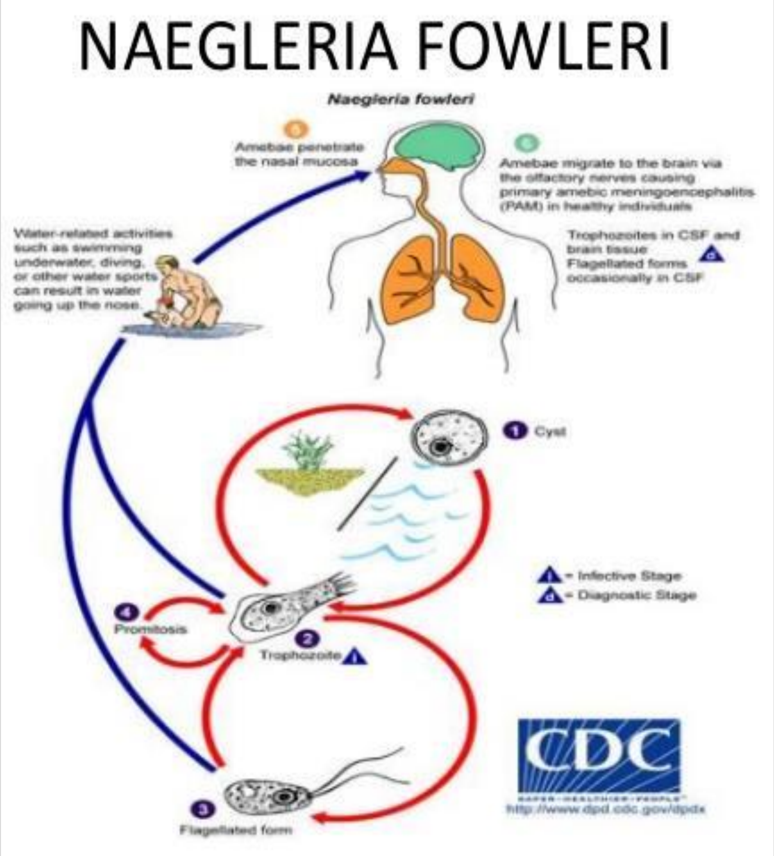
- Blastomycosis, *Candida* spp., paracoccidioidomycosis, phaeohyphomycosis, mucormycosis, actinomycosis, toxoplasmosis, *Sporothrix schenckii*, chromoblastomycoses & *Aspergillus* spp.
- Usually cause abscesses, which may leak into the subarachnoid space to cause chronic meningitis

CNS Infection (Encephalitis)

- Inflammatory process in the brain
- Associated with clinical evidence of neurologic dysfunction
- Most challenging syndromes to diagnose
- Specific pathogen is identified in < 50% of cases
  a) Vast number of infectious causes
  b) Many noninfectious etiologies that present with similar syndromes
  c) Confusion regarding the role of molecular versus serologic testing
- Parasitic: less commonly identified, but are neurotropic & cause sporadic encephalitis and myelitis

Parasitic encephalitis

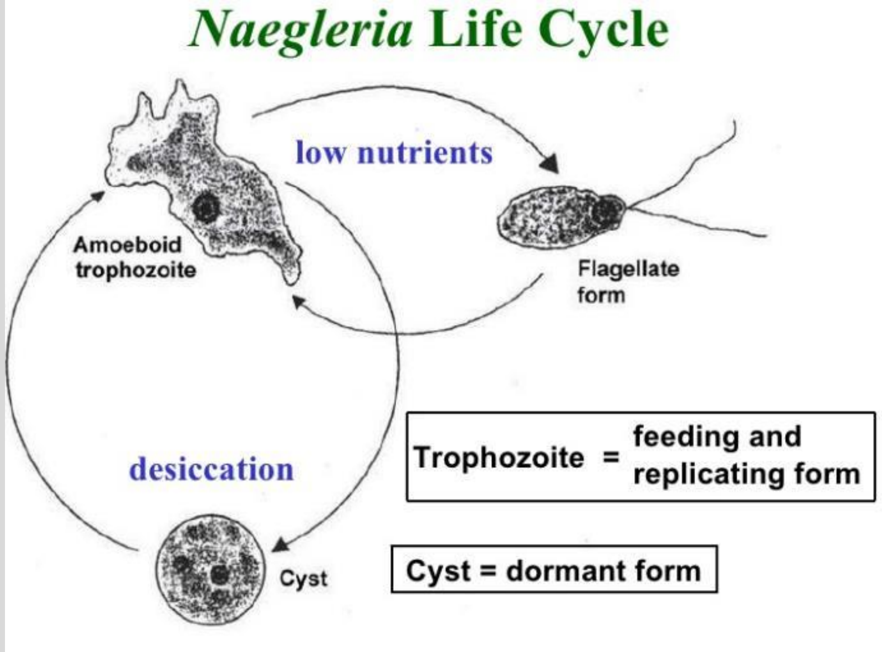
- *Naegleria fowleri*
- Protozoan
- Amebic meningoencephalitis (PAM; Brain-eating disease)
- 100% mortality
- Entry: nose while swimming (not drinking)
- In US: freshwater located in southern-tier states
- Thermophile (46^{o}C)

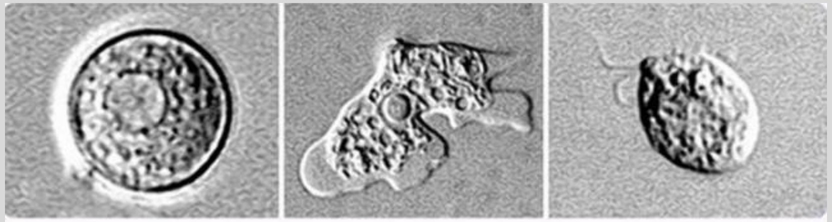


Parasitic encephalitis (*N. fowleri*)

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

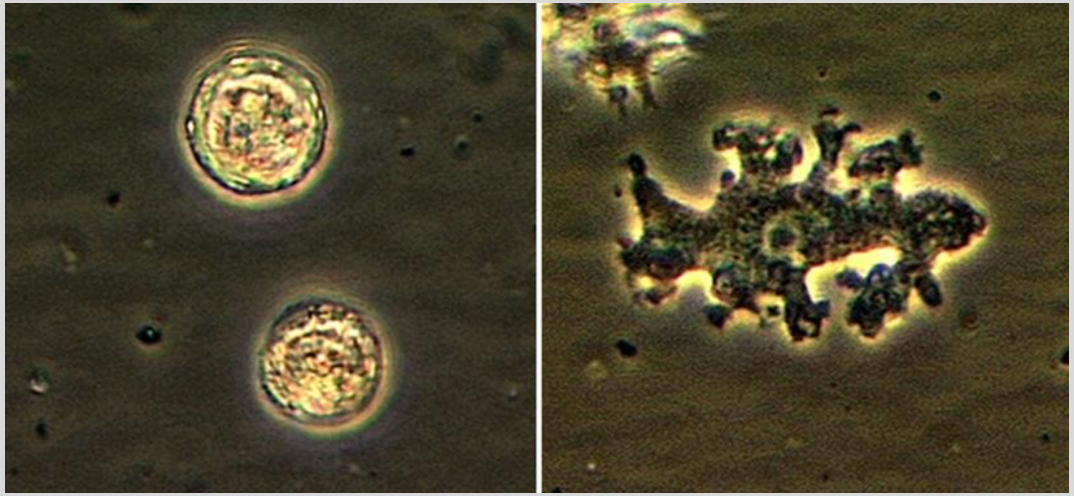
- Symptoms: altered level of consciousness, loss of the sense of smell
- Laboratory:
  a) Neutrophilic pleocytosis
  b) Motile trophozoites on wet mount of warm CSF
  c) Brain histopathology
- Epidemiology:
  a) Summer months
  b) Children & adolescent boys at highest risk
  c) Swimming in fresh water, & particularly water sports
- Treatment:
  a) High mortality even with treatment
  b) Miltefosine shows promise





Parasitic encephalitis

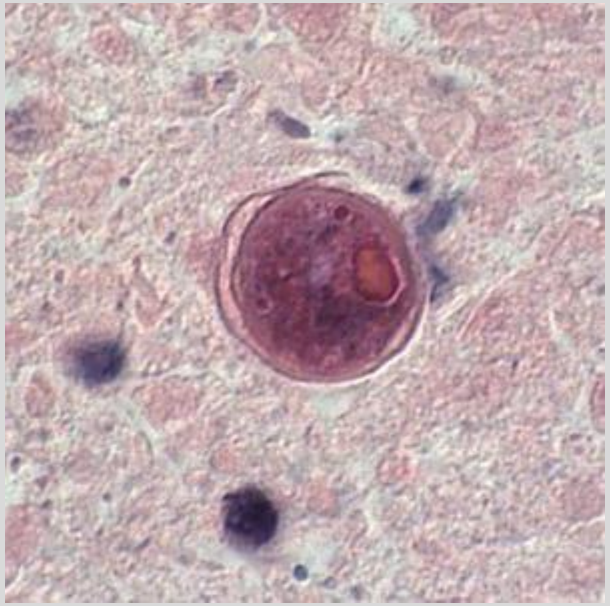
- *Balamuthia mandrillaris*
- Protozoan in soil
- Endemic: Central America
- Cuts and wounds on skin or inhalation
- Presentation (similar to TB; Granulomatous Amebic Encephalitis):
  a) Subacute progressive disease with Space-occupying lesions
  b) Cranial nerve palsies
  c) Hydrocephalus
- Laboratory:
  – Serology (research laboratories)
  – Brain histopathology



https://www.cdc.gov/parasites/balamuthia/index.html

Parasitic encephalitis (*B. mandrillaris*)

- Mortality and Prognosis:
  a) Poor prognosis
  b) Slow progressive disease (2 years)
  c) 89% mortality rate
- Laboratory:
  a) Lymphocyte pleocytosis and depressed glucose (confused with TB)
  b) Serology (Indirect immunofluorescence antibody)
  c) Brain histopathology
  d) Real-time PCR of brain tissue
- Treatment:
  a) Sulfadiazine, a macrolide antibiotic (azithromycin or clarithromycin), rifampin, & fluconazole

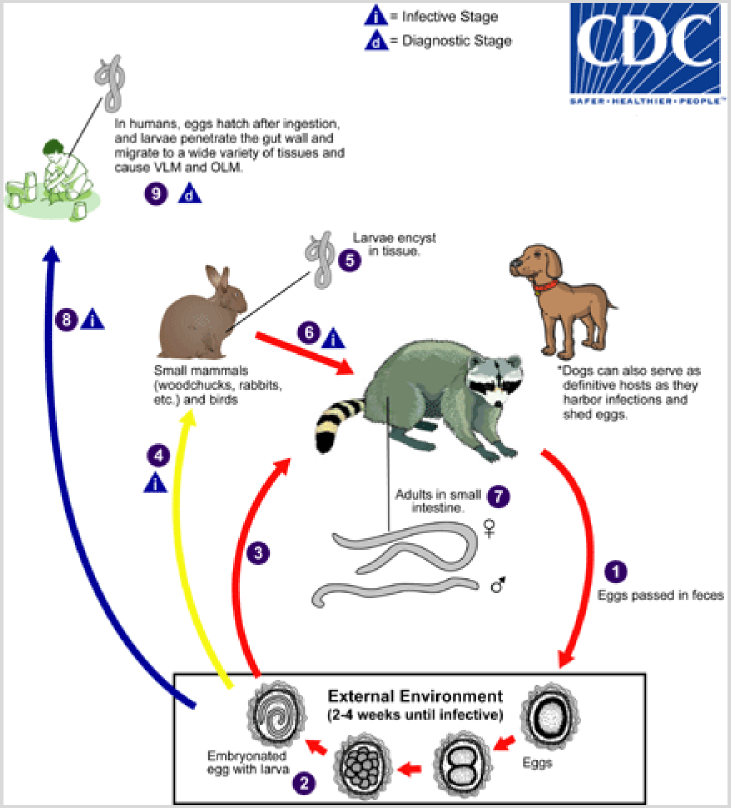


Cyst of *B. mandrillaris* in brain tissue

Parasitic encephalitis

- *Baylisascaris procyonis*
- Raccoon roundworm (helminthes)
- Ingesting infectious eggs (accidental ingestion of dust or dirt)
- Pica/geophagia
- Laboratory:
    a) Eosinophilic pleocytosis
    b) Peripheral eosinophilia
    c) Deep white matter abnormalities (MRI)
    d) Serology on CSF/serum (CDC)
- Epidemiology: Infected raccoons in the mid-Atlantic, northeastern & Midwestern states & parts of California
- Treatment: albendazole

Left: Embryonated egg, showing the developing larva inside. Right: Larva hatching from an egg
https://www.cdc.gov/parasites/baylisascaris/



Parasitic encephalitis

- *Gnathostoma* spp. (round worm)
- Eating undercooked or raw freshwater fish, eels, frogs, birds, & reptiles
- Primarily in tropical and subtropical areas (Thailand, Japan)
- Moves under skin to brain
- Diagnosis:
  a) Microscopic ID of larvae in tissue
  b) ELISA
- Treatment:
  a) Albendazole & ivermectin for skin infection
  b) None for CNS infection

https://www.cdc.gov/parasites/gnathostoma/index.html

Brain abscesses

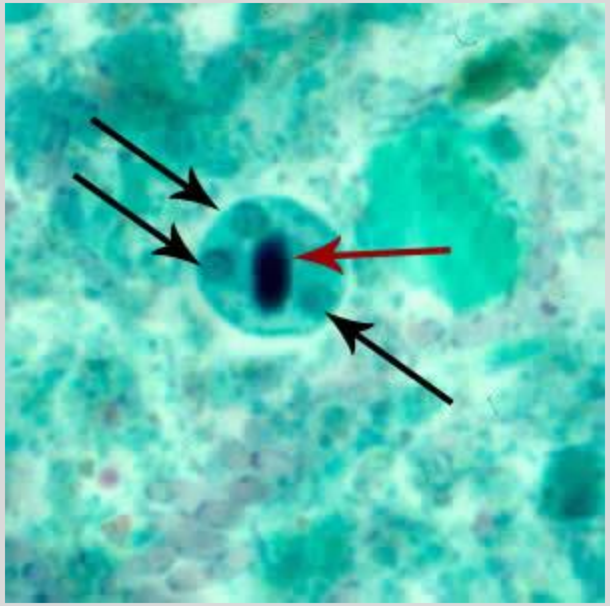
- 2-3 x more common in males then females
- More common in young children and adults over 60
- Predisposing condition: Otitis media or mastoiditis
- Mostly bacterial in etiology
- Parasites that can cause brain abscess:
  a) *Taenia solium* (cysticercosis)*Entamoeba histolytica*
  b) *Schistosoma japonicum*
  c) *Paragonimus* spp.
- Fungi when immunocompromised
  a) *Aspergillus* spp. are especially common in patients who have bone marrow & transplants
  b) T-cell immunity defects (including AIDS patients) are predisposed to *Cryptococcus neoformans*
  c) Neutropenia patients predisposed to *Aspergillus*, *Mucor,* & *Candida*

Brain abscesses

- Clinical manifestations are largely due to presence of space-occupying lesions
- Most common symptoms:
  a) Headache (75% of patients)
  b) Altered mental status
  c) Focal neurologic findings (especially hemiparesis) (>60%)
  d) Fever (50%)
  e) Nausea & vomiting (25–50%)
  f) Seizures (generalized; 30%)
  g) Other signs/symptoms depending on the abscess stage, size & anatomic location
- Labs:
  a) Peripheral leukocytosis (40% normal)
  b) Low serum Na^+ (ADH)
  c) Platelet counts may be high or low
  d) Elevated erythrocyte sedimentation rate
  e) Elevated CRP

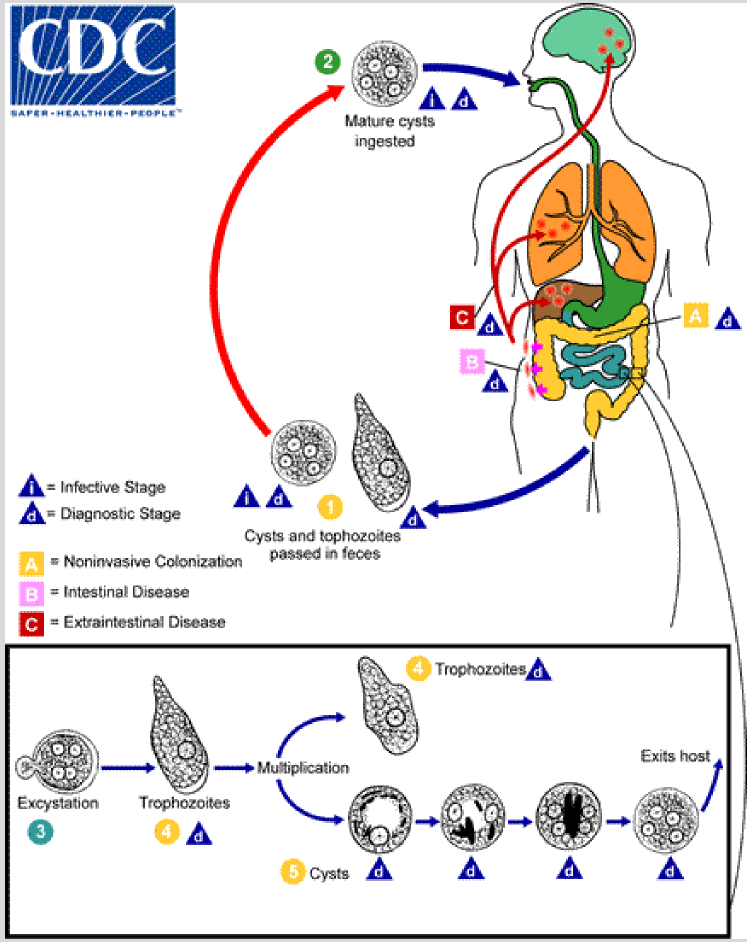
Entamoeba histolytica

- Predominantly a GI disease
- Diagnosis: serology & PCR of brain abscess aspirate or CSF
- Treatment:
  - Metronidazole or tinidazole
  - Followed by paromomycin or iodoquinol



Cyst stained with trichrome. 3 nuclei are visible (black arrows), and the cyst contains a chromatoid body with typically blunted ends (red arrow)

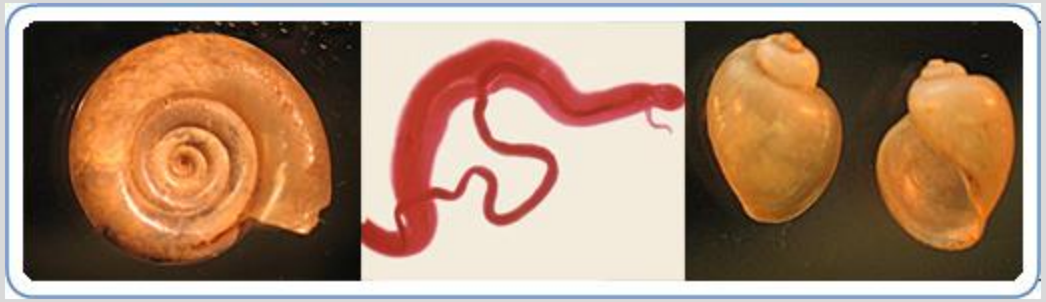
Trophozoite with ingested erythrocytes



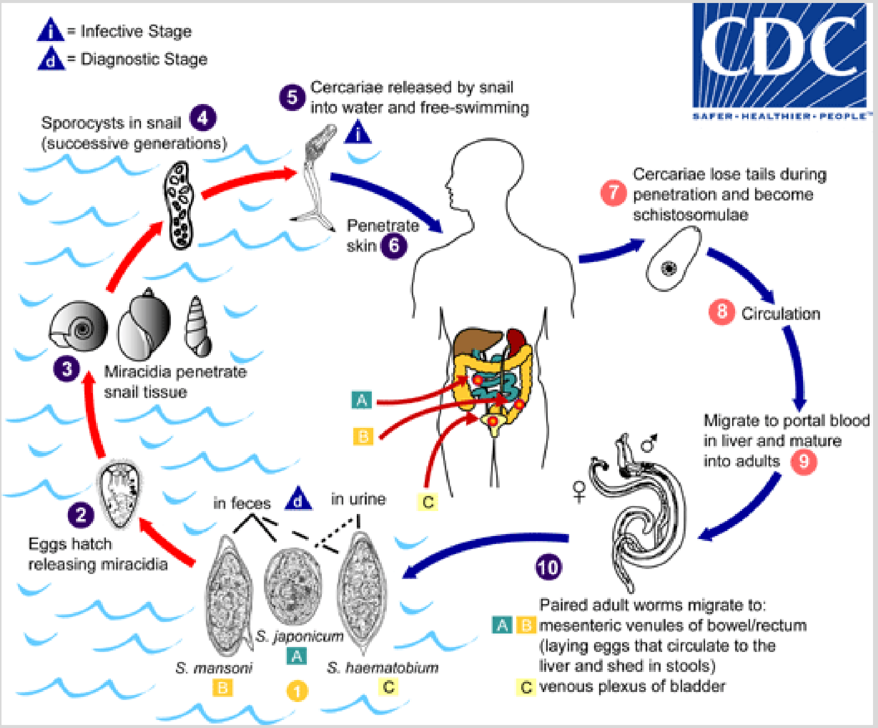
Schistosoma japonicum

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- A.k.a. bilharzia
- Blood fluke
- Not found in the US (careful review of travel & residence history)
- 2nd in impact after malaria (> 200 million)
- Most devastating parasitic disease
- Lives in freshwater snails (skin)
- *S. japonicum* most pathogenic spp. (produce up to 3,000 eggs/day)
- Laboratory:
  - a) Microscopic examination of eggs
  - b) ELISA for anti-schistosomal Abs
  - c) Granuloma formation in brain
- Treatment: praziquantel



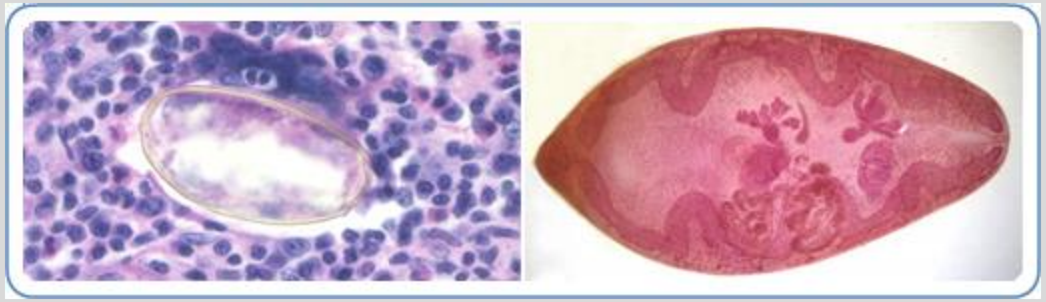
https://www.cdc.gov/parasites/schistosomiasis/index.html



Paragonimus spp.

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Lung fluke (flat worm)
- Ingestion of raw or undercooked crab or crayfish
- Rare in US (multiple cases in Midwest; eating raw crayfish on river raft trips in Missouri)
- *P. kellicotti* (midwestern & southern US in crayfish)
- Symptoms similar to TB
- Laboratory:
  a) Microscopic examination of eggs
  b) Peripheral eosinophilia
- Treatment: praziquantel



Left: Eggs from a lung biopsy stained with H&E. Right: Adult fluke

https://www.cdc.gov/parasites/paragonimus/index.html

TE

A 23 year old female presents to the ER with seizures, confusion and a severe headache. She has recently been touring south america as part of a soccer team. CT scan showed calcified regions in brain parenchyma. Which of the following is the etiological agent of this disease?

- Angiostrongylus cantonensis
- Entamoeba histolytica
- Taenia solium
- Sporothrix schenckii

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

A 23 year old female presents to the ER with seizures, confusion and a severe headache. She has recently been to Thailand as part of a soccer team tournament. The etiological agent was identified with ELISA. Which of the following is the likely source o f this disease?

- Eating raw eels
- Eating raw shell fish
- Aspirating contaminated water
- Ingestion of dirt
- Cuts & wounds on skin

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

TE

A 28-year-old male presents to the ER with symptoms of headache, mental confusion, fever and vomiting. His also felt that his left arm was paralyzed. He is currently a US resident but was born and grow up in Egypt. The etiological agent was confirmed by microscopic analysis of the stool which identified the pathogen eggs. Which of the following is the treatment for this disease?

- Metronidazole
- Praziquantel
- Albendazole
- Ivermectin

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Summary Slide

- Described the different types of fungal and parasitic CNS infections and their etiology
- Compared and contrasted parasitic and fungal meningitis
- Compared the properties of the parasitic and fungal etiological agents of meningitis, encephalitis, myelitis and brain abscesses
- Identified the importance of diagnostic CSF analysis and antimicrobial therapy
- Illustrated epidemiological factors in the determination of etiology

References

- Core References:

1. J. Cohen, S. M. Opal and W. G. Powderly. 2010. **Infectious Diseases**. 3rd Ed. Elsevier Publication. ISBN: 978-0-323-04579-7
2. Jawetz, Melnick, & Adelberg's 2019. **Medical Microbiology** 28th ed.; McGraw Hill Publication. A Lange Medical Book. Parts of Ch. 12, 13, 14 and 20
3. Tao, LE, V. Bhushan and M. Sochat. 2025. **First Aid for the USMLE Step 1**; McGraw Hill Publication.

Lecture Feedback Form:

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine